CogNet Framework

Master Specification — v2.1

Algorithm-First Memory, Belief, Reasoning, and Agent Integration Architecture

Revision objective

This document preserves the core direction of CogNet v2 while integrating the architectural changes required to make the system implementable, model-agnostic, auditable, bounded in compute, and less dependent on LLM intelligence. The central revision is structural: CogNet is no longer treated as a linear nine-layer pipeline. It is organized as four interacting planes governed by an explicit Cognitive Control Plane.

Status: Phase-0 candidate specification. The Minimum Viable Cognition milestone defined here should be implemented and benchmarked before advanced curiosity, reinforcement learning, automatic procedural abstraction, or large-scale multi-agent features.

Version 2.1 — July 2026

Executive Summary

CogNet is a model-agnostic cognitive substrate for agentic systems. It stores typed memories, maintains evidence-backed beliefs, retrieves through bounded activation, reasons over explicit inference structures, tracks contradictions and uncertainty, and returns compact context or actions to any compatible agent.

> **The governing principle remains: algorithm first; model only on unresolved entropy. Deterministic parsing, graph algorithms, local embeddings, small classifiers, explicit evidence rules, and bounded search should do as much work as possible. An LLM is an optional disambiguator and language interface, not the owner of memory or reasoning.**

v2.1 integrates ten major changes

- Cognitive Control Plane: compute budgets, scheduling, stopping rules, escalation, and session state.
- Memory–belief separation: stored propositions are not the same thing as current beliefs.
- Scopes and possible worlds: global, private, project, session, hypothetical, counterfactual, agent-local, and shared-team contexts.
- Causal admission policy: causal edges cannot be learned from co-activation.
- Kernel ontology plus extensions: a stable minimal core with typed specializations.
- Anti-memory-poisoning: trust domains, taint tracking, promotion gates, and mutation rights.
- Negative evidence and failed-search traces: bounded anti-priors against repeated dead ends.
- Typed uncertainty: epistemic, aleatoric, ambiguity, conflict, staleness, and missing-evidence dimensions.
- Provenance DAG: auditable derivation chains from source to decision.
- Associative vs inference graph split: activation discovers candidates; inference structures justify conclusions.

The v2 nine algorithmic layers remain useful as subsystems, but they now live inside four interacting planes: Memory, Reasoning, Epistemic, and Integration/I-O. A Cognitive Control Plane coordinates all four.

Part I — Preserved Foundational Framework

The foundational intent from v2 is preserved: knowledge is not stored as isolated chunks; it is organized as typed, interconnected memory structures. Repeated useful interaction can strengthen accessibility, unused associations can decay, activation can spread through related concepts, and clusters can emerge from use. However, v2.1 makes explicit that frequency, accessibility, truth, utility, and belief are separate quantities.

Core principles retained

- Associative organization rather than isolated storage.
- Activation as a bounded retrieval resource rather than an unbounded database query.
- Reinforcement based on validated usefulness rather than raw recurrence.
- Forgetting as controlled access decay, structural decay, archival, and selective pruning.
- Epistemic tracking so exposure is not mistaken for understanding.
- Emergent clustering through graph structure and usage.
- Universal integration so any agent can use CogNet through a stable protocol.

Critical clarification

A high-scoring associative path is not automatically a valid argument. v2.1 therefore separates candidate discovery from inferential justification. This is one of the most important invariants in the system.

Part II — v2.1 System Architecture

2.1 Architectural Model: Four Planes + Control

```
+------------------------------------------------------+
|               COGNITIVE CONTROL PLANE                 |
| budgets | scheduling | stopping | goals | escalation|
+--------------------------+---------------------------+
                           |
          +----------------+----------------+
          |                |                |
          v                v                v
+----------------+ +----------------+ +----------------+
| MEMORY PLANE   | | REASONING      | | EPISTEMIC     |
| typed IR       | | PLANE          | | PLANE          |
| episodes       | | search         | | evidence       |
| associations   | | rules          | | belief         |
| activation     | | planning       | | conflict       |
| consolidation  | | hypotheses     | | uncertainty    |
+--------+-------+ +--------+-------+ +--------+-------+
         |                  |                  |
         +------------------+------------------+
                            v
+------------------------------------------------------+
|                 INTEGRATION / I-O PLANE               |
| agents | MCP | SDK | tools | docs | code | sensors   |
+------------------------------------------------------+
```

The former nine layers are retained as internal subsystem responsibilities rather than a rigid sequential pipeline. Runtime execution is event-driven and controller-mediated.

3. Cognitive Control Plane

The control plane is Layer 0 in execution terms. It owns reasoning budgets, subsystem scheduling, stopping conditions, escalation policy, and session state. No reasoning loop may execute without a finite budget and explicit termination criteria.

3.1 Budget model

```
B_total = B_graph + B_embed + B_contradiction
        + B_hypothesis + B_tool + B_model

Choose next action:
a* = argmax_a  E[Delta U(a)] / (Cost(a) + epsilon)
```

Delta U is expected utility gain: uncertainty reduction, goal progress, contradiction resolution, or evidence gain. Cost can represent latency, graph expansions, token use, tool calls, energy, or money.

3.2 Required controller state

```
ControlState {
  session_id
  active_goal_ids[]
  remaining_budget
  step_count
  visited_reasoning_states[]
  pending_operations[]
  escalation_level
  termination_reason?
  last_progress_score
}
```

3.3 Mandatory stopping rules

- Budget exhausted.
- Goal confidence crosses a configured threshold.
- Expected information gain falls below threshold.
- No meaningful progress for N controller steps.
- Reasoning state repeats beyond loop tolerance.
- External action is required before further progress.
- Safety or permission boundary blocks continuation.

3.4 Escalation ladder

```
0 deterministic rules / exact index
1 NLP parser / symbolic transforms
2 embeddings / ANN / graph algorithms
3 small local classifier
4 external tool or domain solver
5 LLM disambiguation only when unresolved entropy justifies cost
```

4. Memory Plane: Typed IR and Ontology

4.1 Kernel ontology

v2.1 replaces a globally frozen flat list of memory types with a small stable kernel and extensible specializations.

Kernel type	Purpose	Example specializations
ENTITY	Identity-bearing object	Person, Device, Organization, ConceptEntity
PROPOSITION	Truth-evaluable content	Claim, Hypothesis, Preference
EVENT	Temporally situated occurrence	Episode, Action, StateTransition
PROCEDURE	Executable or reusable process	Plan, Skill, Workflow
GOAL	Desired state / optimization target	TaskGoal, LongTermGoal, Constraint
EVIDENCE	Support or challenge material	Observation, Measurement, TestResult
SOURCE	Origin and trust boundary	User, Document, Tool, Sensor, WebSource

Extensions may define domain-specific types but must inherit lifecycle, provenance, scope, and permission semantics from a kernel type.

4.2 Common Memory IR

```
MemoryObject {
  id: UUID
  kernel_type
  subtype
  scope_id
  payload
  created_at
  valid_time?
  transaction_time
  source_id
  provenance_node_id
  trust_domain
  taint_state
  access_state: HOT|WARM|COLD|ARCHIVED|PRUNED
  embedding_ref?
  counters {
    exposure
    retrieval
    successful_use
    failed_use
    verification
  }
}
```

4.3 JSON boundary rule

JSON is the interchange format. Internal computation should use rigid typed structures, indexed tables, graph adjacency structures, and specialized vector indexes as appropriate. The core kernel must not depend on provider-specific SDK objects.

5. Scopes, Contexts, and Possible Worlds

Every memory, belief, inference, and procedure belongs to an explicit scope. This prevents hypothetical, private, session-local, or agent-local information from contaminating canonical world knowledge.

Scope	Meaning	Default visibility
GLOBAL_WORLD	Canonical shared world model	Policy controlled
USER_PRIVATE	User-specific memory	User + authorized agents
PROJECT	Project-local facts and procedures	Project members
SESSION	Temporary interaction state	Current session
HYPOTHETICAL	Assumed world for reasoning	Reasoning branch
COUNTERFACTUAL	Explicit contrary-to-fact branch	Reasoning branch
AGENT_LOCAL	Private agent state	Owning agent
SHARED_TEAM	Capability-shared multi-agent state	Authorized team

Scope promotion must be explicit. A session observation does not become global knowledge merely because it was repeatedly retrieved.

6. Memory–Belief Separation

Stored content and current belief are different objects. Memory records preserve what was observed, asserted, imported, or derived. Belief states are computed over propositions using current evidence, context, trust, contradiction pressure, and temporal validity.

```
Source -> Evidence -> Proposition -> BeliefState

Memory record:
```

```
  "Sensor voltage was 2.3 V"

Proposition:
  voltage(sensor_x, t1) = 2.3 V

Evidence:
  observation_91 SUPPORTS proposition_17

BeliefState:
  belief(proposition_17 | evidence, scope, time) = 0.84
```

Belief updates must not rewrite source history. New evidence changes the belief state while preserving the complete derivation record.

6.1 Multi-objective memory value

```
Value(m) = [
  truth_confidence,
  empirical_utility,
  accessibility,
  goal_relevance
]
```

These dimensions must not be collapsed prematurely. A useful heuristic may have high empirical utility but low truth confidence; it remains a heuristic or procedure, not a promoted factual belief.

7. Provenance DAG and Memory Integrity

7.1 Provenance DAG

Flat provenance arrays are insufficient for derived knowledge. CogNet must maintain a directed acyclic provenance graph that can trace any conclusion back to raw sources.

```
raw source
   |
   v
observation / imported assertion
   |
   v
normalized proposition
   |
   v
derived proposition
   |
   v
hypothesis
   |
   v
decision / action
```

Each derivation edge records operation, algorithm version, parameters, timestamp, and input object IDs.

7.2 Anti-memory-poisoning architecture

- Every source belongs to a trust domain.
- Every ingested object carries source identity, scope, taint state, and mutation rights.
- Untrusted content may be retrievable without being promotable.
- Untrusted evidence cannot silently modify verified beliefs or executable procedures.
- Promotion requires configurable evidence, corroboration, or authorization gates.
- Imported instructions are data unless explicitly authorized as executable policy.
- Cross-scope propagation requires capability checks.
- High-connectivity growth from a single source is rate-limited and audited.

7.3 Mutation rights

Agents submit observations, evidence, proposed links, feedback, and requested actions. CogNet owns canonical mutation. Direct agent writes to verified beliefs, trusted procedures, or global causal edges are prohibited by default.

8. Input Compilation and Resolution

The deterministic-first fallback hierarchy from v2 is retained. Whole sentences are not fundamental memory units. Input is decomposed into clauses, entities, predicates, temporal relations, negation, modality, and candidate atomic objects.

8.1 Multi-signal resolver

```
R_j = w_s*S_semantic
    + w_l*S_lexical
    + w_n*S_neighborhood
    + w_t*S_type
    + w_c*S_context
    + w_tau*S_temporal
    + w_scope*S_scope
```

Decisions: merge above T_merge; preserve ambiguity in the middle region; create new identity below T_new. Delayed commitment is mandatory. Scope compatibility is added in v2.1 because identical labels can legitimately refer to different local entities.

9. Associative Graph vs Inference Graph

CogNet maintains distinct structures for discovery and justification.

Structure	Purpose	Typical edges / mechanisms
Associative Graph	Find relevant candidates	SIMILAR_TO, CO_ACTIVATED_WITH, SEMANTIC_ASSOCIATION, activation wei
Inference Graph	Justify conclusions	SUPPORTS, CONTRADICTS, DERIVED_FROM, typed rules, causal relations

Invariant: activation discovers; inference validates. A path may be excellent for retrieval and invalid as an argument.

9.1 Edge vocabulary

Core relations include SEMANTIC_ASSOCIATION, CO_ACTIVATED_WITH, SIMILAR_TO, SUPPORTS, CONTRADICTS, PART_OF, INSTANCE_OF, PRECEDES, FOLLOWS, USED_FOR, DERIVED_FROM, OBSERVED_IN, PREDICTS, ENABLES, PREVENTS, and CAUSES.

9.2 Weight and confidence remain separate

```
weight     = cognitive accessibility / traversal preference
confidence = degree of justification

Repeated falsehood:
  weight=0.99, confidence=0.03

Verified theorem seen once:
  weight=0.18, confidence=0.999
```

10. Causal Admission Policy

A CAUSES edge must never be created from co-activation alone. Correlation, sequence, semantic similarity, and repeated co-occurrence are insufficient.

A causal relation may be admitted only with one or more explicit grounds:

- Explicit sourced causal assertion with tracked provenance and source reliability.
- Intervention or controlled experimental evidence.

- Validated domain rule or mechanistic model.
- Temporal precedence plus additional validated causal criteria.
- A hypothesis edge clearly marked provisional, never canonical CAUSES.
- Derived causal inference whose rule and premises are auditable in the provenance DAG.

Causal status

```
ASSOCIATED
CAUSAL_HYPOTHESIS
CAUSAL_SUPPORTED
CAUSAL_VERIFIED
CAUSAL_CONTESTED
CAUSAL_REJECTED
```

11. Bounded Activation and Working Memory

The activation-energy idea is retained and made controller-bounded.

```
Initial:
A_i(0) = alpha*S_i + beta*C_i + gamma*G_i
       + delta*T_i + rho*Scope_i

Propagation:
A_j(t+1) = lambda*A_j(t)
         + sum_i [A_i(t)*W_ij*R_ij*E_j*(1-I_j)]

Lateral inhibition:
I_j = sigmoid(k*(A_competitor - A_j))
```

Only top-K items enter working memory, normally 5–20 objects depending on task class and budget. Candidate priority:

```
P_i = w_a*A_i + w_g*G_i + w_u*U_i
    + w_r*Recency_i - w_c*Cost_i
```

Loop prevention uses visited-state signatures, activation damping, edge revisit penalties, and controller-level no-progress termination.

12. Reasoning Plane

12.1 Explicit reasoning state

```
ReasoningState {
  goal
  subgoals[]
  known[]
  candidate_hypotheses[]
  contradictions[]
  unknowns[]
  frontier[]
  evidence[]
  failed_paths[]
  scope_id
  budget_remaining
}
```

12.2 Query-sensitive algorithms

Query intent	Primary mode
What is X?	Definitional retrieval
Why did X happen?	Backward causal search
What happens if X?	Forward causal search
How do I achieve Y?	Backward planning search
Are X and Y related?	Bidirectional path search

Query intent	Primary mode
What contradicts X?	Contradiction retrieval
What am I missing?	Structural-hole / frontier analysis

12.3 Candidate path score

```
S(P) = sum_i log(W_i + epsilon)
     + alpha*E(P)
     + beta*G(P)
     + gamma*D(P)
     - delta*L(P)
     - eta*C(P)
     - zeta*DeadEndPrior(P)
```

Search may use beam search, A*, bidirectional search, random walk with restart, personalized PageRank, or Monte Carlo sampling. The controller chooses based on intent and budget.

13. Hypothesis Engine and Failed-Path Memory

Hypotheses are provisional propositions generated from activated causes, rules, analogies, or model outputs. Approximate Bayesian ranking is acceptable initially:

```
P(h | E) proportional to P(E | h) * P(h)
```

The next test or observation should maximize expected information gain where practical:

```
IG(test) = H(Hypotheses)
         - E[ H(Hypotheses | result) ]
```

13.1 Negative evidence and dead-end traces

CogNet records bounded failure traces when promising paths repeatedly fail verification in a particular context. These traces create contextual anti-priors; they do not globally ban a path.

```
FailedPathTrace {
  path_signature
  scope_id
  query_intent
  failure_reason
  verification_count
  last_failed_at
  anti_prior_strength
  decay_rate
}
```

Anti-priors decay and are bypassable when materially new evidence appears. This prevents endless rediscovery of the same dead end without creating permanent blind spots.

14. Epistemic Plane

14.1 Epistemic vector

```
EpistemicState {
  confidence
  support
  source_reliability
  cross_context_consistency
  contradiction_pressure
  reasoning_success
  verification_level
  uncertainty {
    epistemic
    aleatoric
    ambiguity
    conflict
    staleness
    missing_evidence
  }
}
```

Typed uncertainty is operational: ambiguity may trigger disambiguation; conflict triggers contradiction analysis; staleness triggers refresh; missing evidence triggers information seeking; aleatoric uncertainty may be irreducible.

14.2 Epistemic states

UNKNOWN, SPECULATIVE, PLAUSIBLE, SUPPORTED, WELL_SUPPORTED, VERIFIED, CONTRADICTED, STALE.

14.3 Contradiction classes

DIRECT_LOGICAL, TEMPORAL, NUMERIC, SOURCE_DISAGREEMENT, ONTOLOGICAL, CONTEXTUAL, APPARENT.

Claims are not silently overwritten. Contradiction analysis first checks scope and valid-time overlap because apparently conflicting claims may both be true in different contexts or periods.

15. Learning, Reinforcement, Forgetting, Consolidation

15.1 Usefulness-gated reinforcement

```
Delta W_ij = eta*A_i*A_j*U*V - lambda*W_ij
```

U is downstream usefulness and V is validation. Retrieval alone does not justify reinforcement.

15.2 Three decay dimensions

```
D = D_access + D_confidence + D_structural
```

- Access decay: retrieval becomes less likely.
- Confidence decay: only for time-sensitive or refresh-sensitive beliefs.
- Structural decay: weak accidental associations weaken or disappear.

Memory states: HOT → WARM → COLD → ARCHIVED → PRUNED. Rare but critical procedures should be protected by policy or utility, not frequency.

15.3 Offline consolidation

- Detect duplicates and near-duplicate propositions.
- Run community detection and update cluster summaries.
- Inspect bridge nodes and recompute centrality.
- Detect contradictions and stale beliefs.
- Decay weak structural associations.
- Archive stale episodes.
- Propose, but do not blindly commit, episodic-to-semantic abstractions.
- Propose procedural extraction only after repeated successful evidence.

Advanced automatic abstraction and procedure extraction are deferred beyond Minimum Viable Cognition.

16. Integration / I-O Plane

CogNet is provider-independent. The core reasoning kernel must never import a provider SDK.

16.1 Core API

```
POST /memory/ingest
POST /memory/query
POST /memory/activate
```

```
POST /reason
POST /feedback
POST /observe
POST /goal
POST /contradiction
POST /consolidate
GET  /context
GET  /state
```

16.2 Agent tool surface

```
memory.observe(...)
memory.recall(...)
memory.explore(...)
memory.explain_path(...)
memory.challenge(...)
memory.hypothesize(...)
memory.verify(...)
memory.goal(...)
memory.feedback(...)
memory.forget(...)
```

16.3 First integration target

Recommended: generic MCP-compatible agent integration as the architectural target. Claude Code may be the first demonstration client, but CogNet must not belong to any single model vendor.

16.4 Context contract

```
{
  "goal": {...},
  "working_memory": [],
  "relevant_facts": [],
  "candidate_hypotheses": [],
  "contradictions": [],
  "recommended_procedures": [],
  "unknowns": [],
  "belief_summary": [],
  "typed_uncertainty": {...},
  "budget_state": {...}
}
```

17. Explicit Inspectable Reasoning Protocol

This is an engineered control protocol, not private chain-of-thought. It records inspectable state transitions, evidence references, scores, and decisions.

Step	Operation	Owning subsystem
1	PARSE — classify problem and compile input	Integration + Memory
2	ACTIVATE — seed relevant objects	Memory
3	DECOMPOSE — create subgoals	Reasoning
4	RETRIEVE — gather evidence	Memory
5	GENERATE — candidate paths/hypotheses	Reasoning
6	CHALLENGE — seek contradictions	Epistemic
7	SCORE — rank surviving candidates	Reasoning + Epistemic
8	IDENTIFY GAPS — locate missing evidence	Epistemic
9	SEEK INFORMATION — maximize expected gain	Control + Integration
10	SYNTHESIZE — construct supported conclusion	Reasoning
11	CALIBRATE — attach typed uncertainty	Epistemic
12	ACT — choose response/tool/action	Control + Integration
13	OBSERVE — capture outcome	Integration + Memory

Step	Operation	Owning subsystem
14	LEARN — update utility/failure evidence	Memory + Epistemic
15	CONSOLIDATE — long-term restructuring	Memory

The controller may skip irrelevant steps, repeat bounded search/hypothesis cycles, or terminate early. Every loop consumes budget and is subject to no-progress detection.

18. Minimum Viable Cognition (MVC)

v2.1 narrows the first serious implementation target. The goal is not to build every research feature. The goal is to prove that a weak model can gain measurable memory-dependent capability from an algorithmic cognitive substrate.

MVC includes

- Rigid typed Memory IR with kernel ontology.
- Provenance DAG.
- Scopes and possible worlds.
- Associative graph.
- Belief layer separate from memory.
- Bounded activation and top-K working memory.
- Deterministic contradiction detection.
- Typed uncertainty.
- One causal reasoning mode with causal admission policy.
- Failed-path traces and contextual anti-priors.
- Cognitive Control Plane with budgets and stopping rules.
- MCP-compatible integration surface.
- Benchmark harness against no-memory, vector RAG, and summary baselines.

Explicitly excluded from MVC

- Curiosity engine and self-directed exploration.
- Full reinforcement learning.
- Automatic procedure extraction.
- Automatic episodic-to-semantic abstraction.
- Large-scale multi-agent shared cognition.
- Distributed graph database.
- Biologically realistic neuron simulation.
- Fancy 3D embedding visualization.
- Custom LLM training.
- Unrestricted LLM-driven graph rewriting.

19. Revised Implementation Roadmap

Phase	Target	Deliverable / exit condition
0	Freeze v2.1 spec	Versioned schemas, scopes, permissions, provenance, belief model, control budgets, benchmark definitions.
1	Deterministic kernel	Typed IR, CRUD, SQLite, graph traversal, provenance DAG, scopes, tests. No LLM.
2	Resolver	Local embeddings, ANN, multi-signal resolution, ambiguity states; measured merge/split accuracy.
3	Activation + control	Budgeted activation, inhibition, top-K workspace, stopping rules, loop prevention.
4	Belief + epistemics	Evidence graph, belief computation, typed uncertainty, temporal validity, contradiction rules.
5	MVC reasoning	One causal mode, path scoring, hypotheses, failed-path traces, explicit uncertainty.
6	Integration	Python SDK, HTTP API, MCP adapter, context compiler; provider-independent kernel.
7	Benchmark gate	A/B/C/D evaluation: no memory, vector RAG, summary, CogNet. Publish failure analysis.
8	Learning loop	Outcome feedback, usefulness reinforcement, anti-feedback-loop controls, counterfactual tests.
9	Consolidation	Communities, archival, merge proposals, guarded abstraction.
10	Research extensions	Curiosity, active evidence seeking, procedural extraction, advanced planning.

Roadmap rule: no phase advances solely because code exists. Each phase needs measurable exit criteria and regression tests.

20. Phase-0 Decisions — Recommended Defaults

Decision	v2.1 default
Memory domain	Arbitrary observations: conversations, documents, code, tools, web evidence, sensors; normalized through adapters.
Sharing	Yes, but scoped and capability-based; no giant communal brain.
World vs personal	Strictly separated scopes.
Mutation rights	Agents propose/observe; CogNet owns canonical mutation.
Contradictory models	Yes, through contexts and possible worlds.
Causality	Separate from association; governed by causal admission policy.
Privacy	Local-first core with optional sync.
First integration	Generic MCP architecture; Claude Code may be first demo client.
Internal representation	Rigid typed IR internally; JSON at boundaries.
Truth vs usefulness	Track separately; never collapse to one objective by default.

21. Benchmarks and Falsification

CogNet is a research claim only if it can lose. Benchmarks must be designed to falsify the architecture, not merely showcase it.

Core metrics

- Retrieval Precision@K, Recall@K, MRR.
- Entity resolution accuracy and merge/split error.
- Contradiction detection precision, recall, F1.
- Path relevance and inference validity as separate metrics.
- Causal false-positive rate.
- Memory growth rate and graph density over time.
- Query latency and compute budget consumption.
- Token reduction and LLM-size degradation.

- Long-horizon consistency and recovery after failure.
- Catastrophic association rate and stale-memory error rate.
- Poisoning resistance and unauthorized promotion rate.
- Belief calibration: Brier score / expected calibration error where applicable.
- Repeated-dead-end rate before and after failed-path memory.

Primary experiment

```
Same task suite + same weak LLM:

A. no memory
B. vector RAG
C. conversation summary
D. CogNet MVC

Measure:
success, tokens, latency, consistency,
recovery, personalization, calibration,
poisoning resistance, long-horizon retention
```

Headline comparison

```
Strong LLM + basic RAG
          versus
Weak LLM + CogNet
```

The architecture should be considered successful only where improvements survive controlled baselines and ablations. The result is empirical, not predetermined.

22. Non-Negotiable System Invariants

- The LLM is never the canonical owner of memory mutation.
- No unbounded reasoning loop.
- No causal edge from co-activation alone.
- No silent overwrite of contradictory claims.
- No promotion across scopes without policy.
- No inference conclusion without traceable evidence or explicitly marked speculation.
- No conflation of accessibility, truth confidence, empirical utility, and goal relevance.
- No assumption that an associative path is a proof.
- No destructive forgetting solely because a memory is rare.
- No provider SDK dependency inside the core kernel.
- No high-trust promotion from untrusted content without gates.
- No benchmark claim without baseline and ablation.

23. Closing Framing

CogNet v2.1 is not a vector database with a graph-shaped interface, and it is not an attempt to simulate consciousness. It is a model-agnostic, uncertainty-aware, self-organizing cognitive substrate that performs memory selection, scoped association, evidence tracking, belief maintenance, contradiction management, bounded reasoning, and auditable integration outside the LLM.

The central engineering bet is now precise: a weak language model should be able to perform materially better on memory-dependent agentic tasks because CogNet supplies structured memory, bounded search, explicit belief state, contradiction handling, and evidence-aware reasoning.

The next implementation milestone is Phase 0 freeze followed by the Minimum Viable Cognition path. Advanced curiosity, autonomous abstraction, and richer learning should be treated as research extensions only after the MVC survives benchmark comparison.

End of CogNet Framework Master Specification v2.1